TANG CHEE HENG

Smart Contract Engineer | DeFi Protocol Researcher

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SUMMARY

Blockchain engineer and DeFi researcher specializing in protocol architecture, yield strategies, and smart contract development. Built security-audited core smart contracts powering Polystream, an ERC-4626 multi-chain yield protocol integrating 35+ DeFi yield strategies. Experienced in Foundry-based testing, smart contract security, DeFi protocol analysis, on-chain yield mechanisms, and building full-stack Web3 products across mobile, web platforms, and decentralized applications, with a background in artificial intelligence and machine learning. Seeking roles in smart contracts development, smart contract audits, and protocol infrastructure.

PROFESSIONAL EXPERIENCE

Co-founder, Protocol Engineer, Polystream **Feb 2025 - Present**

- Co-founded Polystream, a DeFi yield aggregation protocol originating from the Scroll Open Hackathon (2nd place, $15K USD in prize), selected for the Scroll Campus accelerator program, and later securing $95k USD in Scroll Foundation investments and grants.
- Designed and built the core Solidity smart contracts powering an ERC-4626 vault architecture supporting multiple assets across 3 blockchain networks.
- Researched and evaluated DeFi protocols and on-chain yield strategies by reviewing protocol architectures and smart contract implementations, analyzing permission models, capital flows, and potential security risks to ensure safe integration.
- Integrated 35+ DeFi protocols and yield strategies cross-chain vault architecture, engineering allocation mechanisms that allows capital distribution across strategies to target 8–12% APY.
- Implemented safeguarded curator-controlled strategy execution with whitelisted and guarded smart contract interactions, performing security reviews of integrated DeFi protocols to assess permission models, external function calls, security assumptions, and potential attack vectors prior to integration.
- Developed Foundry-based testing infrastructure including unit, integration, invariant, and fuzz testing to detect accounting inconsistencies and potential security vulnerabilities in vault logic.
- Built backend infrastructure including event listeners, oracle integrations, monitoring services, and vault trackers for real-time vault management.
- Built full-stack applications including a web dApp, mobile DeFi app, and curator dashboard for vault monitoring and strategy management.
- Led the security audit process with Cantina for Polystream's core smart contracts, working with auditors to analyze findings including potential OOG DoS vectors, vault accounting edge cases, and precision loss in exchange rate, implementing mitigation fixes and contract improvements.

Fullstack AI Intern, Vilor Berhad **June 2024 - Sep 2024**

- Developed the Vilor internal intranet platform, enabling centralized document management, HR workflows, and operational tools used across finance, HR, and administration teams.
- Implemented HR process automation including dynamic form-to-document generation (Word, Excel, PDF) supporting 19 HR workflow templates and 31 job description templates, improving HR workflow efficiency by 20% and reducing document generation errors by 70%.
- Designed internal systems for document storage, form submission, and company-wide information management, including fiscal calendars, company holidays, and centralized reference resources for employees.
- Processed and analyzed 13 million WhatsApp sales advisory messages using Python, extracting 2,770 frequently asked customer questions to construct a training dataset and knowledge base for onboarding new sales staff.
- Built an AI chatbot prototype for sales and customer service training by fine-tuning a large language model on constructed knowledge base derived from internal communication data.

PROJECTS

Decentralized Credit Scoring for DeFi Lending, Final Year Project

- Analyzed 2,000+ on-chain wallets, extracting behavioral features from transactions, asset holdings, and smart contract interactions to construct a DeFi credit scoring dataset.
- Trained CatBoost, Gradient Boosting, and Random Forest models, achieving ~98% prediction accuracy (R^2 up to 0.985) for wallet credit risk assessment.
- Built a semi-supervised learning pipeline in Python and integrated Chainlink External Adapters + Chainlink Node to submit and store credit scores on-chain via smart contracts.
- Developed a full-stack Web3 dApp allowing users to submit wallet addresses and retrieve generated credit scores

DeESG – Decentralized ESG Scoring Platform, ETH GLOBAL BANGKOK

- Winner – Chainlink "Connect the World" ($2K), Innovation on Scroll 5th Place ($500).
- Built a decentralized ESG scoring platform using AI clustering and IoT environmental data from Raspberry Pi sensors.
- Implemented Chainlink oracles, external adapters, and Chainlink Automation to push verified IoT data on-chain for ESG evaluation.
- Developed smart contracts and token incentives to store ESG scores and reward verified environmental data, integrating FHE-based anonymous governance forums.

AI Guardian – Deepfake & Image Verification, ETH KL

- Winner – ETH KL Best Community Application on Scroll ($1K), ICP–Ethereum Chain Fusion Challenge ($1K).
- Built an AI-powered deepfake detection platform with on-chain verification using ICP Canisters and World ID.
- Developed smart contracts to store image hashes and classify images as authentic or deepfake.
- Integrated AI inference, blockchain storage, and frontend systems for real-time verification.

TECHNICAL SKILLS

Blockchain & Smart Contracts

Solidity, Foundry (Unit, Integration, Invariant & Fuzz Testing), Hardhat, Ethers.js, Wagmi, Chainlink (Data Feeds, Automation, Data Streams, Node), Pyth, RedStone, Account Abstraction, Smart Accounts, Bundlers, Paymasters

DeFi & On-chain Research Tools

DeFiLlama, Dune Analytics, DeBank, Etherscan, Tenderly, The Graph, Moralis, Infura, Alchemy

Backend & APIs

Node.js, Express, FastAPI, Docker

AI / Machine Learning

Python, Scikit-learn, CatBoost, XGBoost, K-Means, FAISS, LangChain, HuggingFace, OpenAI

Frontend

React, Next.js, Vue.js, React Native, TypeScript, JavaScript, Tailwind CSS

Databases

PostgreSQL, TimescaleDB, MongoDB, Firebase, SQL

Developer Tools

Git, Docker, PM2, VSCode, K6

EDUCATION

2021-2025

University of Nottingham

BSc Computer Science with Artificial Intelligence

2016-2020

QS World Ranking: Top 100

First Class Honours with average result of 79%

Dean's List Scholarship Award Top 2%(2022, 2024)

SMJK JIT SIN

SPM: 10A (5A+, 4A, 1A-)